

Task Manager V6 Full Design Specification

Purpose

V6 introduces Search, Filter, and Sort capabilities. The goal is to make larger task lists easier to navigate without changing how tasks are stored. Unlike V5, V6 does not change task fields, constructors, or file formats.

Version Goal

Add Search, Filter, and Sort while preserving all V1-V5 functionality.

Why V6 Exists

After V5 added priority and category, users need ways to find and organize tasks more efficiently. V6 solves that problem through views of existing task data.

Architecture Decisions

Main handles menus, user input, and displaying results. TaskManager handles search, filter, and sort logic. Task stores task data. Maintain Main -> TaskManager -> Task architecture.

Search Design

Search only checks task names. Search does not use category, priority, or completion status because filtering already handles those.

Search Rules

Partial matching. Case-insensitive. Display-only. Example: searching 'java' finds Study Java, Java Homework, and Learn Java OOP.

Search Validation

No tasks available message when task list is empty. No matching tasks found message when nothing matches. Blank searches are not allowed.

Filter Design

Single Filter menu containing Filter by Priority, Filter by Category, and Filter by Completion Status.

Priority Filter

High, Medium, and Low priority filtering.

Category Filter

School, Work, Personal, Fitness, Church, and Other category filtering.

Completion Status Filter

Completed and Incomplete filtering.

Filter Behavior

Filtering is display-only. It never modifies, deletes, or reorders tasks.

Sort Design

Single Sort menu containing Sort by Priority and Sort by Completion Status.

Priority Sorting

Order is High -> Medium -> Low because the most important tasks should appear first.

Completion Sorting

Order is Completed -> Incomplete because the project owner preferred seeing accomplishments first.

Sort Behavior

Sorting is display-only. Sort a copied list and leave the original order unchanged.

TaskManager Methods

searchTasks(), filterByPriority(), filterByCategory(), filterByCompletionStatus(), sortByPriority(), sortByCompletionStatus().

Return Types

All search, filter, and sort methods return ArrayList. TaskManager provides data and Main handles presentation.

Reusable Display Method

Create displayTasks(ArrayList) and use it for View, Search, Filter, and Sort results.

Menu Layout

1 Add Task, 2 View Tasks, 3 Search Tasks, 4 Filter Tasks, 5 Sort Tasks, 6 Complete Task, 7 Delete Task, 8 Rename Task, 9 Mark Task Incomplete, 10 Set Task Priority, 11 Set Task Category, 12 Exit.

Result Formatting

All Search, Filter, and Sort results use the same numbered task format used by View Tasks.

File Format

No file format changes. Continue using name,completed,priority,category.

Testing Checklist

Test search matching, search validation, filters, sort order, original order preservation, and all V1-V5 features.

V6 Complete When

Search works, Filter works, Sort works, results are numbered, original order is unchanged, save/load still works, and all previous features still work.

Future Version Notes

V7 will likely focus on refactoring, reducing duplication, helper methods, and architecture cleanup.